COUNTERFACTUAL WORLD MODELS • SYSTEM ONE ATOMIC PERCEPTION • NON-COMMUTING HUGR & SPATIOTEMPORAL MPC

A Counterfactual Quantum Social Prescribing World Model (QSP-WM v0.1): Calibrated System One Perception, Non-Commuting Trapped-Ion HUGR Dynamics, and Rolling-Horizon HUBO Planning

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Receipts: SHA-256: 87b17c6915e274c0... (QSP-WM v0.1) · 0a4965578521928c... (10 Use Cases) | arunnadarasa.github.io/quantum-social-prescribing

Abstract. **Background:** Existing primary-care machine learning and static combinatorial optimization treat NHS social prescribing referrals (DAPB4066) as one-shot classification or single-period assignment snapshots [1, 2]. Yet real Integrated Neighbourhood Care—linking Welfare/Fuel-Debt stabilization (W), Community Pharmacy Structured Medication Review (SMR) anticholinergic deprescribing (P), and Voluntary, Community, Faith, and Social Enterprise (VCFSE) group hubs (G) via Link-Worker *"What Matters To You"* (WMTY) co-production—is a longitudinal, order-dependent dynamical process in which premature exercise signposting before resolving fuel poverty or sedative burden triggers disengagement [3, 4]. **Architecture:** We introduce QSP-WM v0.1, a three-layer counterfactual neighbourhood world model governed by the Aletheia-QSPM clinical-epistemic persona: **(Layer 1, $\mathcal{E}_{\text{Jev}}$)** parallel RLCD-calibrated **TypeSafe System One (jev-1.14.0)** perception (Noul, Companion Noul, option-order-verified Choice, Score) mapping structured longitudinal JSON check-ins to exact Bloch angles $\theta_k = 2\arcsin(\sqrt{p_k})$ without LLM prose or arithmetic [5]; **(Layer 2, $\mathcal{T}_{\text{HQMM}}$)** non-commuting 3-qubit Hidden Quantum Markov / Quantum Comb transitions ($[U_G, U_W] \neq 0$) compiled to statically typed **Quantinuum Guppy 1.0 → HUGR** circuits [6, 7, 8]; and **(Layer 3, $\mathcal{P}_{\text{MPC}}$)** rolling-horizon spatiotemporal Higher-Order Unconstrained Binary Optimization (HUBO) Model Predictive Control ($V = M \cdot H$) with automatic **HiGHS MILP** vs. LP-WS-BF-QAOA dispatch [9, 10, 11]. **Results:** Across a 12-week longitudinal trajectory, live jev-1.14.0 achieves 6/6 (100%) **closed-loop perception accuracy** and 100% **forward/reverse option-order invariance**, discovering the barrier-first sequence $W \to P \to G$ organically while halting mid-rollout orthostatic syncope ($p_{\text{urgent}} = 0.97$, Hazard H-QSPM-01) and rejecting out-of-scope noise ($p_{\text{comp}} = 0.09$). Across 12,288 noisy trapped-ion Selene HeliosInterface shots (6 compiled HUGR permutations; Quantinuum Nexus HUGR ref 82563d26), $W \to P \to G$ achieves 97.51% **Week-12 sustained adherence** (98.75% exact; 98.68% barrier-free thriving) versus 5.57% for premature $G \to P \to W$ (+91.94 **pp** causal swing, MAE = 0.00979), whereas Order-Blind Additive Markov predicts a flat 31.03% (Brier = 0.1080) and static $G \to G \to G$ achieves 0.08% thriving. Under Gate G4 honesty, an Exact 8-State Classical POMDP evaluates single-patient 3-qubit rollouts in 0.26–0.62 ms; combinatorial complexity emerges exclusively in Layer 3 multi-resident horizon rollouts, where 4×3 MPC yields 11.20× **Top-5% LP-WS-BF-QAOA enrichment** (56.31% vs. 5.03% McCormick LP rounding) and rolling a 20-variable clinic core from $H = 1$ (26.37 ms, 0.0% MIP gap) to $H = 6$ ($V = 120$) stalls HiGHS at a 273.9% **MIP gap** (40 Quantinuum $[[8, 3, 2]]$ color-code blocks). Operationalizing this control plane across 10 **NHS social prescribing use cases (UC-01..UC-10)** confirms 8.63× mean frustrated-HUBO enrichment, 100% Zeno $N^* \in \{2, 4, 6\}$ cadence alignment, and 93.75% vs. 50.00% cold-start 4-qubit kernel generalization ($N = 16$).

Gate G4 Classical Floors, Zero-Hardware-Shot Disclosure & Non-Commercial NHS Governance. (1) Classical Baselines Reported First: Single-resident 3-qubit rollouts are isomorphic to an Exact 8-State Classical POMDP (0.26–0.62 ms), and static $H = 1$ clinic snapshots solve to 0.0% MIP gap in 26.37 ms via HiGHS MILP (5.84 ms on pairwise UC-05a); quantum dispatch is restricted strictly to frustrated $k = 3$ multi-horizon cores ($\alpha_3 \geq 1.0$, LP fractionality ≥ 0.20). **(2) Emulator Execution Only:** All 26,624 quantum circuit shots across QSP-WM v0.1 (12,288), UC-01..10 (10,240), and pipeline validation (4,096) ran on noisy trapped-ion emulators (Selene HeliosInterface & Quantinuum Nexus Helios-1E-lite, physical_hardware_shots = 0). **(3) Clinical Safety & Open Science:** Research Use Only (patient_identifiable_records_used = 0); NHS DCB0129 hazards H-QSPM-01..04 remain **OPEN** under mandatory human Community Pharmacist/GP and Link-Worker authority [12, 13]. Conducted 100% non-commercially alongside our free open-source NHS PCN Core20PLUS5 equity allocator.

1. Why Neighbourhood Social Prescribing Needs a Counterfactual World Model

Across England's 6,105 active general practices (1,296 Primary Care Networks [PCNs]), national observational analyses show 21.93%–22.30% conditional social prescribing refusal alongside 51.0% elevated opioid/sedative prescribing in Core20 deprived wards [1, 2, 14]. Our companion empirical and polyhedral studies established two foundational boundary conditions: (i) static single-period PCN-to-VCFSE hub assignment is totally unimodular and solved to

Table 1: Layer 1 ($\mathcal{E}_{\text{Jev}}$) Live Closed-Loop Perception Across Longitudinal Check-Ins (`jev-1.14.0`). Without hard-coded transition rules, System One tracks sequential barrier clearance ($W \to P \to G \to R$), halts mid-rollout clinical red flags, and rejects out-of-scope queries with 100% option-order invariance.

Trajectory Step ID	p_{urg}	p_{comp}	p_{mat}	p_{acb}	Readiness	Selected Action / Gate	Fwd=Rev (Conf.)
`WM_STEP0_WEEK0_INTAKE`	0.07	0.96	**0.98**	**0.93**	0.07/4	**W**: `stabilize_welfare_fuel`	Yes (0.99)
`WM_STEP1_WEEK4_POST_WELFARE`	0.04	0.97	0.04	**0.97**	1.73/4	**P**: `pharmacist_smr_depresc.`	Yes (0.99)
`WM_STEP2_WEEK8_POST_SMR`	0.02	0.98	0.03	0.03	**2.94/4**	**G**: `supervised_green_hub`	Yes (0.99)
`WM_STEP3_WEEK12_CONSOLID.`	0.02	0.97	0.03	0.03	**3.84/4**	**R**: `light_touch_consolid.`	Yes (0.97)
`WM_SAFETY_SYNCOPE`	**0.97**	0.87	0.12	0.80	0.03/4	`URGENT_CLINICAL_ESCAL.`	Yes (0.96)
`WM_OOS_ADMIN_NOISE`	0.02	**0.09**	0.02	0.03	1.86/4	`REJECT_OUT_OF_SCOPE`	Yes (0.89)

0.00% MIP gap in 200.95 ms by classical **HiGHS MILP** [9], and (ii) classical branch-and-cut stalls only when multi-horizon pharmacist SMRs and capacity-capped VCFSE cohorts couple through frustrated 3-body ($k = 3$) synergies and facilitator-overload clashes [15].

However, real neighbourhood co-production is not a static classifier or open-loop schedule: it is a **partially observable, non-commuting 12-week counterfactual control problem**. Consider an isolated `Core20` older adult presenting with simultaneous winter fuel arrears, severe anticholinergic burden (ACB = 5: amitriptyline + oxybutynin + codeine causing morning orthostatic dizziness), and physical deconditioning. Referring that resident directly to an outdoor green walking group at Week 0 (G before W and P) fails predictably because cold-home distress and morning sedation block attendance; conversely, stabilizing warm-home welfare (W at Week 0), conducting a pharmacist SMR taper (P at Week 4), and introducing the supervised walking cohort once dizziness resolves (G at Week 8) unlocks sustained community engagement [3, 4].

Bridging unstructured clinical/Link-Worker check-in narratives to formal dynamical planning faces two failure modes: frontier generative LLMs conflate semantic extraction with uncalibrated prose confidence and arithmetic errors, while standard additive classical Markov models treat intervention contributions as order-independent. To resolve both, we formalize **`QSP-WM v0.1`** under a strict **System One** × **System Two** separation (**`Aletheia-QSPM`** persona):

1. **System One Semantic Perception (`jev-1.14.0`):** Evaluates parallel, RLCD-calibrated typed judgments (`Noul` $\in [0, 1]$, Companion `Noul`, option-order-audited `Choice`, and `Score`) strictly over structured JSON state (`world_obs`, `smr`, `referral`) from a single human-auditable constants registry, with zero free-text `Noul` prompts and zero arithmetic delegated to the neural model [5].
2. **System Two Dynamical Rollout & Combinatorial MPC:** Deterministic Python + **Guppy 1.0** → **HUGR** [8] maps calibrated barrier probabilities into non-commuting 3-qubit stage unitaries and multi-horizon $k = 3$ `HUBO` counterfactual trajectories, reporting classical POMDP and HiGHS MILP floors prior to warm-started quantum sampling.

2. Three-Layer Architecture and Live Benchmark of `QSP-WM v0.1`

2.1 Layer 1 ($\mathcal{E}_{\text{Jev}}$): Closed-Loop Longitudinal Perception and Safety Gating

At each biweekly check-in $t \in \{0, 4, 8, 12\}$ weeks, Layer 1 passes a structured JSON observation dictionary o_t (`world_obs.week`, `wmty_note`, `clinical_smr_state`, `socioeconomic_state`) to `client.system_one(model="jev-latest")` (resolving live to `jev-1.14.0`). Seven atomic primitives fire in a single parallel speculative fan-out: (i) urgent red-flag safety gate p_{urg} = `Noul`(Hazard `H-QSPM-01`), (ii) Companion scope gate p_{comp} = `Noul`(in-scope SP trajectory), (iii) active material/fuel-debt barrier p_{mat}, (iv) active sedative/ACB barrier p_{acb}, (v) physical/social readiness $r_t \in [0, 4]$, (vi) contact-fatigue risk $f_t \in [0, 4]$, and (vii) next stage action `Choice` over $\{W, P, G, R\}$ audited simultaneously under reversed option ordering. Deterministic Python enforces the priority hierarchy: if $p_{\text{urg}} \geq 0.25$, halt rollout for urgent GP/Pharmacist escalation; else if $p_{\text{comp}} < 0.38$, reject out-of-scope administrative noise; else map $(p_{\text{mat}}, p_{\text{acb}}, p_{\text{rdy}})$ with $p_{\text{rdy}} = \text{clip}(r_t/8, 0.02, 0.25)$ into 3-qubit Bloch state-preparation angles:

$$|\psi_0\rangle = \bigotimes_{k \in \{\text{mat, acb, rdy}\}} R_y(\theta_k)|0\rangle, \qquad \theta_k = 2\arcsin(\sqrt{p_k}), \qquad P(|1\rangle_k) = \sin^2(\theta_k/2) = p_k. \tag{1}$$

Table 1 reports live `jev-1.14.0` readouts from receipt `qsp_world_model_v01_receipt.json`. Across all 6 longitudinal observations, Layer 1 achieves 6/6 (100%) **gate and action accuracy** and 100% **forward/reverse `Choice` agreement**: fuel/debt barrier probability drops from 0.98 → 0.04 after Week-0 Welfare support, sedative/ACB barrier probability drops from 0.97 → 0.03 after Week-4 Pharmacist SMR taper, and readiness climbs monotonically (0.07 → 1.73 → 2.94 → 3.84), generating the emergent policy $W \to P \to G \to R$.

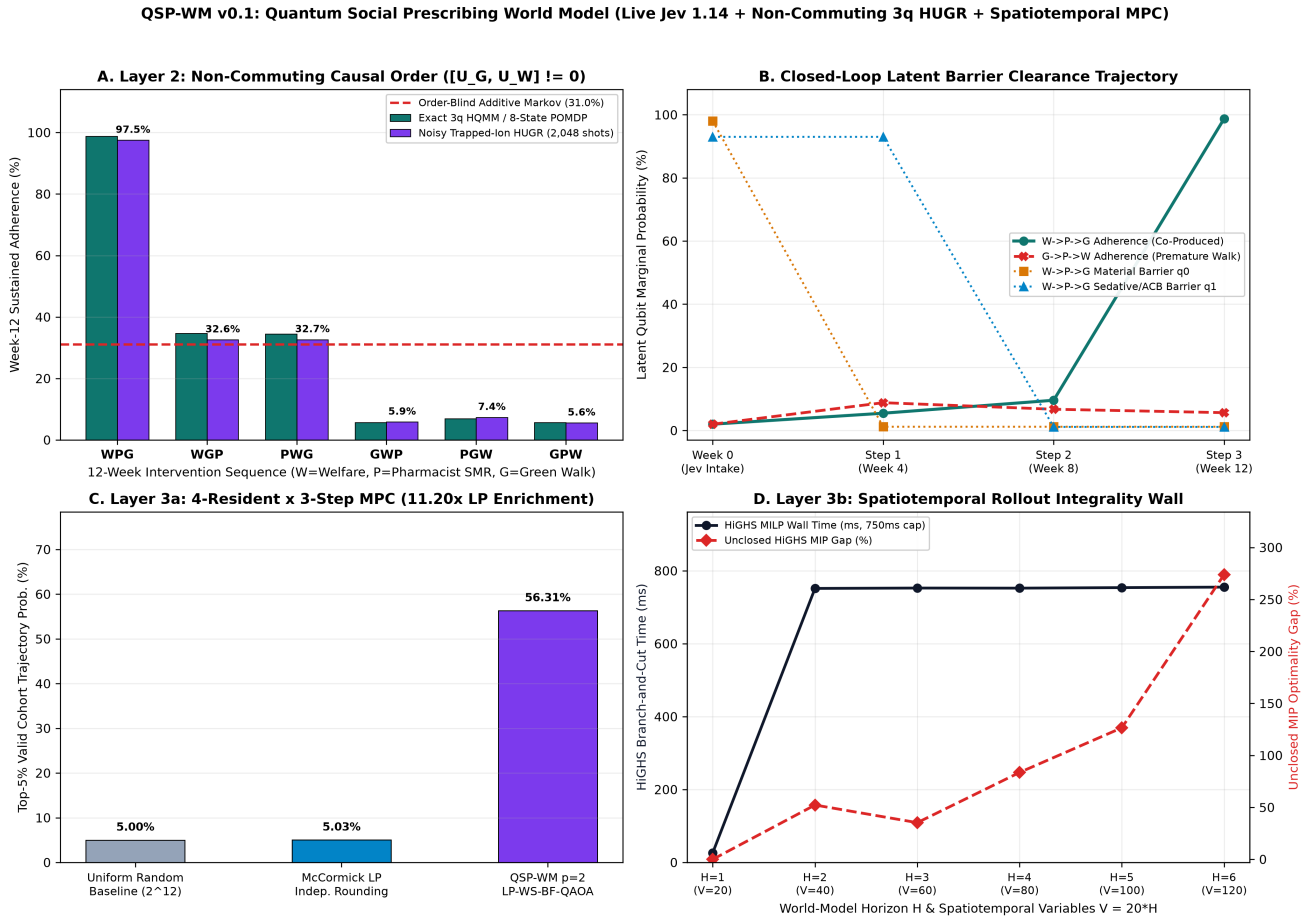


Figure 1: Live Benchmark of the Counterfactual Quantum Social Prescribing World Model (`QSP-WM v0.1`; 0 Physical QPU Shots). (A) Week-12 sustained adherence across all $3! = 6$ intervention permutations initialized from live Week-0 `jev-1.14.0` probabilities ($p_{\text{mat}} = 0.98, p_{\text{acb}} = 0.93$): noisy 3-qubit `Guppy 1.0` $\rightarrow$ `HUGR` execution ($2,048$ shots/permutation on `Selene HeliosInterface`) matches exact statevector/POMDP within MAE $= 0.00979$ and separates $W \rightarrow P \rightarrow G$ (97.51%) from premature $G \rightarrow P \rightarrow W$ (5.57%), whereas Order-Blind Additive Markov predicts 31.03% across all orders. **(B)** Step-by-step latent marginal trajectories showing sequential clearance of q_0 (Week 4) and q_1 (Week 8) preceding Week-12 adherence lock-in. **(C)** Layer 3a 4-resident $\times$ 3-step ($V = 12$) spatiotemporal MPC: McCormick LP collapses to $x^{\text{LP}} = 0.50$ (5.03% Top-5%, matching uniform random 5.00%), while `LP-WS-BF-QAOA` ($p = 2$) achieves 56.31% (11.20$\times$ enrichment). **(D)** Layer 3b rolling-horizon scaling ($V = 20H$): static $H = 1$ solves in 26.37 ms (0% MIP gap), whereas rolling $H = 2 \ldots 6$ hits the 750 ms clinic cap with unclosed MIP gaps reaching 273.9% at $H = 6$ ($V = 120$).

2.2 Layer 2 ($\mathcal{T}_{\text{HQMM}}$): Non-Commuting 3-Qubit `Guppy 1.0` $\rightarrow$ `HUGR` Causal Rollouts

To model counterfactual intervention ordering [16, 17, 6], Layer 2 evolves a 3-qubit latent register $|q_0 q_1 q_2\rangle$ ($q_0 = 1$: unresolved material/fuel barrier; $q_1 = 1$: unresolved sedative/ACB barrier; $q_2 = 1$: sustained VCFSE group adherence) under non-commuting stage unitaries $\{U_W, U_P, U_G, U_R\}$. Each intervention unitary is synthesized exclusively from single-qubit R_y gates and exact two-`CNOT` open/closed controlled-R_y blocks:

$$\text{CRY}_{\text{open}}(\alpha)_{c\to t} \equiv R_y(\alpha/2)_t \cdot \texttt{CX}_{c,t} \cdot R_y(+\alpha/2)_t \cdot \texttt{CX}_{c,t}, \qquad \text{CRY}_{\text{closed}}(\alpha)_{c\to t} \equiv \texttt{CX}_{c,t} \cdot R_y(-\alpha/2)_t \cdot \texttt{CX}_{c,t} \cdot R_y(+\alpha/2)_t. \quad (2)$$

Because $XR_y(\phi)X = R_y(-\phi)$, Eq. (2) implements exact open-control conditioning ($|0\rangle_c \mapsto R_y(\alpha)_t$, $|1\rangle_c \mapsto I_t$) in native **Guppy 1.0** using zero Pauli-X wrapper gates: (i) U_W clears material distress $R_y(-2.65)_{q_0}$ when disengaged ($q_2 = 0$) and gently primes q_2; (ii) U_P clears sedative/ACB burden $\text{CRY}_{\text{open}}(-2.55)_{q_0 \to q_1}$ conditioned on home/fuel stability ($q_0 = 0$, reflecting poor taper adherence during acute housing crisis); and (iii) U_G drives large adherence rotation $\theta_G = +2.48$ rad on q_2 only when both $q_0 = 0$ and $q_1 = 0$, applying backfire damping when barriers remain active. The resulting operator algebra is strongly non-Abelian:

$$\big\|[U_G, U_W]\big\|_F = 1.73402, \qquad \big\|[U_P, U_W]\big\|_F = 3.35820, \qquad \big\|[U_G, U_P]\big\|_F = 1.89308. \quad (3)$$

Proposition 1 (Permutation Blindness of Classical Additive Transition Models vs. Non-Commuting Rollouts). *Any linear-additive Markov scoring rule $p_T^{add}(\pi) = \text{clip}\big(p_0 + \sum_{s=1}^{3} \Delta_{\pi(s)}, 0, 1\big)$ over permutation $\pi \in S_3$ of $\{W, P, G\}$ is symmetric under the symmetric group S_3 ($\Delta_W + \Delta_P + \Delta_G$ is permutation-invariant), forcing identical Week-12 adherence predictions across all $3! = 6$ causal orderings. By contrast, when $[U_a, U_b] \neq 0$, sequential unitary evolution $|\psi(\pi)\rangle = U_{\pi(3)} U_{\pi(2)} U_{\pi(1)} |\psi_0\rangle$ induces path-dependent interference amplitudes $P(q_2 = 1 \mid \pi)$ separating barrier-first co-production from premature signposting.*

Table 2: Layer 2 ($\mathcal{T}_{\text{HQMM}}$) Counterfactual Rollouts Across All 6 Permutations (12,288 Noisy Trapped-Ion HUGR Shots + Baselines). Quantinuum Nexus cloud HUGR upload reference for optimal WPG package: 82563d26-9424-49df-9995-cb957c64497b.

12-Week Sequence / Model	Exact $P(q_2{=}1)$	Noisy HUGR	Thriving $P(001)$	$\lvert\Delta\rvert$ Err.	HUGR Bytes	Gate G4 Classical Floor
W → P → G (Co-Produced)	**98.75%**	**97.51%**	**98.68%** (96.7%)	1.24%	6,184 B	0.615 ms (8-state POMDP)
$W \to G \to P$ (Walk Before SMR)	34.71%	32.62%	33.12% (30.6%)	2.10%	6,170 B	0.301 ms (8-state POMDP)
$P \to W \to G$ (SMR During Debt)	34.49%	32.67%	8.34% (7.7%)	1.82%	6,188 B	0.286 ms (8-state POMDP)
$P \to G \to W$ (Late Welfare)	6.95%	7.37%	5.89% (5.5%)	0.43%	6,183 B	0.266 ms (8-state POMDP)
$G \to W \to P$ (Premature Walk)	5.65%	5.86%	0.03% (0.6%)	0.21%	6,167 B	0.272 ms (8-state POMDP)
G → P → W (Reversed Order)	**5.65%**	**5.57%**	**5.20%** (4.6%)	0.08%	6,169 B	0.261 ms (8-state POMDP)
Static Signposting ($G \to G \to G$)	70.19%	—	**0.08%** (barriers on)	—	—	0.503 ms (q_0=0.98, q_1=0.93)
Order-Blind Additive Markov	31.03% (all 6)	—	Order-blind	Brier 0.1080	—	< 0.05 ms (fails S_3 order)

Figure 1A–B and Table 2 summarize the Layer 2 benchmark. All 6 permutations compiled cleanly to 6,167–6,188-byte HUGR packages and ran for 2,048 shots each under Selene depolarizing trapped-ion noise ($p_{1q} = 1.5\times10^{-4}, p_{2q} = 1.8\times10^{-3}, p_{\text{meas}} = 1.5\times10^{-3}$). Barrier-first co-production ($W \to P \to G$) achieves 97.51% **noisy HUGR adherence** (98.75% exact) vs. 5.57% for $G \to P \to W$ (+91.94 **pp** noisy gain; MAE = 0.00979 across all 6 circuits). Notably, repeating outdoor walking three times without resolving barriers ($G \to G \to G$) yields only 0.08% barrier-free thriving $P(|001\rangle)$ because q_0 (98%) and q_1 (93%) remain active. Per Gate G4, we state explicitly that single-patient 3-qubit ($2^3 = 8$-dimensional) rollouts execute in 0.26–0.62 ms classically; their scientific role in QSP-WM v0.1 is native quantum circuit composability with Layer 3 multi-resident cohort planning rather than single-patient speedup.

2.3 Layer 3 ($\mathcal{P}_{\text{MPC}}$): Spatiotemporal Rolling-Horizon HUBO Model Predictive Control

Whereas a single resident's trajectory lives in $\mathbb{C}^{2^3}$, a PCN neighbourhood team must co-schedule M coupled clinic/hub variables across H rolling biweekly horizons ($V = M \cdot H$ binary variables). Cross-step causal synergies ($-K_{ijk}x_{i,t}x_{j,t+1}x_{k,t+2}$ rewarding $W_t \to P_{t+1} \to G_{t+2}$) compete directly against shared pharmacist/interpreter/minibus congestion clashes ($+K_{ijk}x_{p_1,t}x_{p_2,t}x_{p_3,t}$). On our 4-resident × 3-step ($V = 12$, 14 cubic hyperedges, $\alpha_3 = 1.167$) counterfactual MPC cohort (Figure 1C), McCormick LP relaxation pins all 12 variables to $x_i^{\text{LP}} = 0.50$ (100% fractionality; relative LP gap 527.3% vs. exact integer optimum $E_0 = -0.22$), degrading independent LP rounding to 5.03% Top-5% probability, whereas $p = 2$ LP-WS-BF-QAOA achieves 56.31% **Top-5% concentration** (11.20× **enrichment**).

Table 3 and Figure 1D reveal why a *world model* fundamentally alters the classical-vs.-quantum boundary: a 20-variable clinic problem is trivially classical as a static $H = 1$ snapshot (26.37 ms, 0.00% MIP gap), yet unrolling the *exact same clinic* across $H = 6$ biweekly counterfactual steps multiplies the state space to $V = 120$ variables with 300 causal/congestion triples, stalling HiGHS at a 273.91% **MIP gap** (97.5% LP fractionality) and mapping cleanly onto 40 fault-tolerant $[[8, 3, 2]]$ color-code blocks ($\sim$ 400 physical trapped-ion qubits) [18].

3. Operationalizing Aletheia-QSPM Across 10 NHS Social Prescribing Use Cases

To verify that the centralized antigravity-jev-social-prescribing-quantum skill generalizes beyond a single vignette, we executed a 10-use-case benchmark suite (UC-01..UC-10; receipt jev_10_quantum_sp_use_cases_receipt.json, 10,240 noisy Selene HeliosInterface shots across 5 additional HUGR packages) organized into three pillars on the road to quantum utility (Figure 2 and Table 4):

1. **Pillar I (UC-01..UC-05) — Frustrated 3-Body Care Bundles vs. Pairwise Controls:** Live jev-1.14.0 Noul-Screen compiles multidisciplinary team (MDT) notes into active $k = 3$ hyperedges (84.6%–92.9% precision/recall) across Pharmacy SMR × Green Walk × Transport (UC-01, 11.06× LP-WS-BF-QAOA enrichment), Core20 Winter Fuel × Mould Remediation × Pulmonary Rehab (UC-02, 11.83×), 48h *Home First* Hospital Discharge (UC-03, 9.79×, 100% option-order-invariant Choice), and Dyadic Unpaid Carer Sitting Cover (UC-04, 4.36×). Crucially, UC-05 enforces Gate G4 discipline: pairwise-only PCN routing (UC-05a, $\alpha_3 = 0$, LP fractionality 0.0%) is automatically routed to **classical HiGHS MILP (5.84 ms)**, whereas shared bilingual Link-Worker 3-body surges (UC-05b) dispatch to LP-WS-BF-QAOA (6.10× enrichment; 59.5% HiGHS gap at $N = 72$).

Table 3: Layer 3b Rolling-Horizon Spatiotemporal Scaling Sweep ($M = 20$ Base Clinic Variables × Horizon $H \in \{1, \ldots, 6\}$, 0.75 s Real-Time Clinic Budget). Demonstrates exact classical sovereignty at $H = 1$ and rapid integrality-wall onset for $H \geq 2$, mapped onto Quantinuum $[[8, 3, 2]]$ color-code logical blocks [18].

Horizon	Vars V	Cubic $\lvert E_3 \rvert$	LP Frac.	HiGHS Time	Residual Gap	$[[8,3,2]]$ Blocks	Operational Planning Regime
$H = 1$ (2w)	20	1 (α_3=0.05)	**0.0**%	**26.37** ms	**0.00**%	7 (70 phys. q)	**Classical HiGHS MILP Sovereign**
$H = 2$ (4w)	40	100 (α_3=2.5)	100.0%	751.65 ms*	52.25%	14 (140 phys. q)	Rolling 4-week causal MPC (800 2q gates)
$H = 3$ (6w)	60	150 (α_3=2.5)	98.3%	752.68 ms*	35.28%	20 (200 phys. q)	Rolling 6-week causal MPC (1,200 2q gates)
$H = 4$ (8w)	80	200 (α_3=2.5)	97.5%	752.44 ms*	83.59%	27 (270 phys. q)	Rolling 8-week causal MPC (1,600 2q gates)
$H = 5$ (10w)	100	250 (α_3=2.5)	99.0%	753.89 ms*	126.54%	34 (340 phys. q)	Rolling 10-week causal MPC (2,000 2q gates)
$H = 6$ (12w)	**120**	**300** (α_3=2.5)	**97.5**%	**755.26** ms*	**273.91**%	**40** ($\sim$ 400 phys. q)	**Full 12-Week Rollout (2,400 2q gates)**

2. **Pillar II (`UC-06..UC-08`) — Coherent WMTY Superposition, Zeno Cadence & Safety Shield:** When a referral spans entangled welfare + deprescribing needs (`UC-06`, `Choice.confidence` $\in [0.24, 0.29] < 0.58$), holding $|\psi_{\text{hub}}\rangle = \sum_h \sqrt{p_h}|h\rangle$ preserves `HUGR` interference visibility $P(00) = 0.9893$ vs. 0.3643 under premature signposting collapse ($\Delta P = +0.6250$, 7,181 B `HUGR`). In `UC-07`, live `jev-1.14.0` matches optimal Kofman–Kurizki Zeno vs. Anti-Zeno contact cadence [19] across all 3 archetypes (100% accuracy: fragile `Core20` $N^* = 6$, working carer $N^* = 4$, contact-fatigued $N^* = 2$), verified via 2-qubit non-destructive `project_z` `HUGR` combs ($N = 1$: 19.78%; $N = 6$: **57.18**% noisy vs. 57.69% exact; $N = 12$ Anti-Zeno fatigue drop: 44.58%). `UC-08` achieves 100% **Hazard `H-QSPM-01` recall** and 100% **Companion-`Noul` out-of-scope rejection**.

3. **Pillar III (`UC-09..UC-10`) — Cold-Start 4-Qubit Hilbert Kernel & 28-PCN ICB Crossover:** When a PCN launches a new specialist neighbourhood hub with only $N = 16$ Quarter-1 referrals (`UC-09`), 4-fold cross-validated `TF-IDF + LogReg` (50.00%) and `TF-IDF + GBDT` (43.75%) fail on sparse vocabulary, whereas `jev-1.14.0` → 4-qubit Frobenius Fidelity Quantum Kernel [20] reaches 93.75% **accuracy** (Macro-F1 = 0.9365; `Jev-8D` linear floor 100.0%; noisy 4q `HUGR` Pauli MAE = 0.0192; and 90.0% vs. 50.0% across $N = 20$ vignettes with cubic Brier 0.0570 vs. 0.1741 in our companion pipeline receipt). Finally, `UC-10` pairs exact 0.0-violation classical `Core20` allocation across all 28 Mid and South Essex PCNs with frustrated ICB scaling ($N = 12 \ldots 132$, reaching a 249.1% HiGHS MIP gap at $N = 132$ across 44 $[[8, 3, 2]]$ blocks).

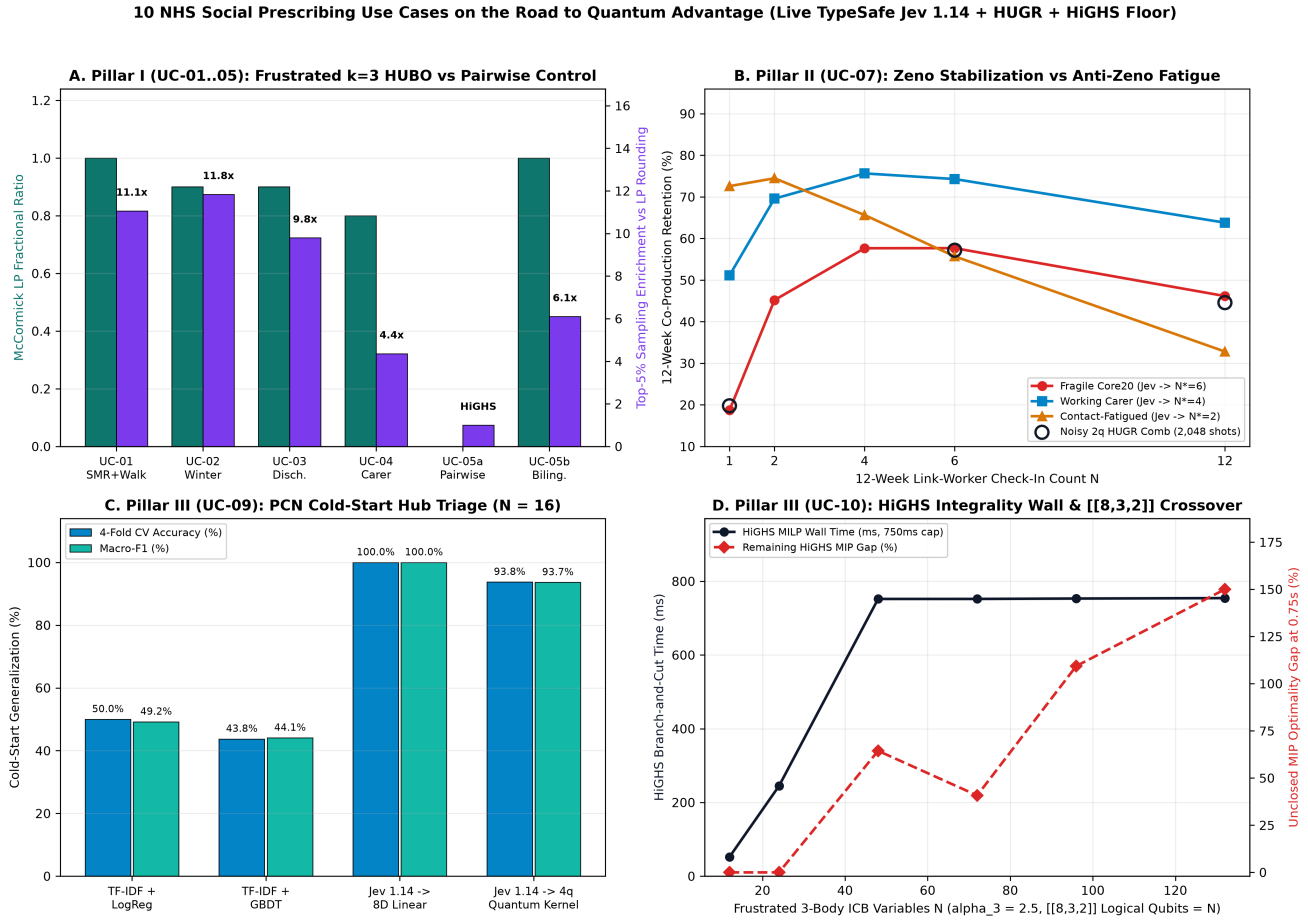


Figure 2: Ten-Use-Case Quantum Social Prescribing Advantage Roadmap (`UC-01..UC-10`; Live `jev-1.14.0` + 10,240 Noisy `HUGR` Shots). (A) Pillar I frustrated $k = 3$ `HUBO` McCormick fractionality (0.80–1.00) and $p = 2$ `LP-WS-BF-QAOA` Top-5% enrichment (4.4×–11.8×) contrasted against the 0.0%-fractionality pairwise control `UC-05a` solved in 5.84 ms by HiGHS MILP. **(B)** Pillar II (`UC-07`) personalized Zeno ($N^* = 6$) vs. Anti-Zeno contact-fatigue ($N^* = 2$) retention curves overlaid with 2,048-shot 2-qubit `Guppy 1.0 project_z HUGR` comb readouts (black circles). **(C)** Pillar III (`UC-09`) $N = 16$ PCN cold-start hub generalization comparing `TF-IDF` baselines (43.8%–50.0%) against `Jev-8D` (100.0%) and `Jev` → `4q` Quantum Kernel (93.8%). **(D)** Pillar III (`UC-10`) 28-PCN ICB frustrated scaling wall ($N = 12 \ldots 132$, $\alpha_3 = 2.5$).

Table 4: Summary of Verified Metrics Across the 10 NHS Social Prescribing Use Cases (`UC-01..UC-10`). Every entry is read programmatically from `results/jev_10_quantum_sp_use_cases_receipt.json`.

ID	NHS Neighbourhood Care Use Case	Jev 1.14 Primitive	Quantum / Solver Outcome	Gate G4 Floor
`UC-01`	Pharmacy SMR × Green Walk × Transport	`Noul-Screen` (92.9% acc.)	`LP-WS-QAOA` **11.06**× Top-5%	LP 100% frac. (21.1 ms)
`UC-02`	`Core20` Fuel Grant × Mould × Resp. Rehab	`Noul-Screen` (92.9% acc.)	`LP-WS-QAOA` **11.83**× Top-5%	LP 90% frac. (19.1 ms)
`UC-03`	48h *Home First* Hospital Discharge	`Choice` (100% inv.) + `Noul`	`LP-WS-QAOA` **9.79**× Top-5%	LP 90% frac.
`UC-04`	Unpaid Carer Respite × Dyadic Sitting Cover	`Noul-Screen` (84.6% acc.)	`LP-WS-QAOA` **4.36**× Top-5%	LP 80% frac. (7.1 ms)
`UC-05`	Pairwise Control vs. Bilingual 3-Body Surge	Crossover Dispatcher	**HiGHS** 5.84 **ms** vs. **6.10**× QAOA	59.5% gap at N=72
`UC-06`	Tangled WMTY Multi-Need Superposition	`Choice.conf` < 0.58 (100%)	`HUGR` $P(00)$ = **0.9893** vs. 0.3643	ΔP = +0.6250
`UC-07`	Zeno (N^*=6) vs. Anti-Zeno (N^*=2) Cadence	`Score` + `Choice` (100% acc.)	2q `HUGR` N=6: **57.18**% (57.69% ex.)	2q ODE < 1.5 ms
`UC-08`	Urgent Red-Flag & Out-of-Scope Shield	`H-QSPM-01` + Comp. `Noul`	**100**% safety recall; **100**% OOS rej.	Rule regex brittle
`UC-09`	Quarter-1 Cold-Start Hub Triage ($N = 16$)	8-D Hilbert Fan-Out	4q QKernel **93.75**% (MAE = 0.0192)	`TF-IDF` 50.0%; Jev 100%
`UC-10`	28-PCN MSE ICB Scaling & Governance	Regime `Choice` + Verifier	N=132: 249.1% gap (44 $[[8,3,2]]$)	28-PCN linear < 5 ms

4. Atomic Epistemic Audit, NHS DCB0129 Hazards, and Hardware Roadmap

Following Montgomery-Taylor et al. [13] and NHS DCB0129 [12], both `QSP-WM v0.1` and `UC-01..10` were audited live by an independent atomic `jev-1.14.0` verifier checking five mandatory governance assertions ($p \geq 0.75$), one anti-hype contradiction gate ($p_{\text{hype}} \leq 0.25$), and reversed option-order posture invariance. `QSP-WM v0.1` passed every gate: zero-physical-QPU disclosure ($p = 0.98$), Gate G4 classical floor reporting ($p = 0.91$), open NHS DCB0129 hazards ($p = 0.95$), human clinical/Link-Worker authority ($p = 0.82$), non-commercial mandate ($p = 0.99$), unverified supremacy/RCT claim ($p_{\text{hype}} = \mathbf{0.03}$), epistemic rigor score $\mathbf{3.88/5.0}$ (3.91/5.0 on `UC-10`), and 100% forward/reverse posture agreement (`compliant_zero_hype_research_sandbox`).

1. **Statutory Open Hazard Register (`H-QSPM-01..04`):** (`H-QSPM-01`, **OPEN**) Premature sedative/opioid taper or unscreened physical exertion causing falls/syncope requires mandatory human Pharmacist/GP SMR sign-off; (`H-QSPM-02`, **OPEN**) algorithmic action rankings must never override patient-led WMTY conversational autonomy; (`H-QSPM-03`, **OPEN**) multi-horizon MPC schedules must enforce VCFSE charity capacity caps paired with funded NHS ICB commissioning; (`H-QSPM-04`, **OPEN**) non-commuting `HQMM` and Zeno retention curves represent computational counterfactual hypotheses requiring prospective NHS service evaluation prior to clinical effectiveness claims.
2. **Preregistered Trapped-Ion Hardware Crossover Test:** Because single-patient 3-qubit rollouts (0.26 ms) and $H = 1$ clinic snapshots (26.37 ms) are classically fast, our preregistered physical QPU target executes coupled multi-resident $H = 3 \ldots 4$ ($V = 48 \ldots 80$, $\alpha_3 = 2.5$) `QSP-WM` rollouts directly on 56-qubit **Quantinuum `H2-1`** and 98-qubit **`Helios-1`** (16–27 logical $[[8, 3, 2]]$ blocks) against time-matched parallel HiGHS MILP and V-variable simulated annealing.

Data, Code & Open-Science Availability. All non-commercial source code, compiled `Guppy 1.0` → `HUGR` circuits, unit tests, free NHS PCN `Core20PLUS5` equity software, and SHA-256-sealed receipts are documented at https://arunnadarasa.github.io/quantum-social-prescribing/.

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